# *Unit - 2*

**Question:**
**Define nanomaterials. Explain their properties.**

**Answer:**

Nanomaterials have an average grain size less than 100 nanometers. "Nano" comes from Greek *nanos* (dwarf); scientifically means $10^{-9}$ (1 nm = $10^{-9}$ m).

**Properties:**

- Nanomaterials lie between atomic/molecular and bulk scale.
- Show different properties from bulk due to:

  – Large surface atom fraction
  – High surface energy
  – Spatial confinement
  – Fewer imperfections

- Very high surface area to volume ratio → enhances surface-related behavior.

**Question:**
**Explain the surface area, optical, and catalytic properties of nanomaterials with examples.**

**Answer:**

<u>Surface Area:</u>

- As particle size reduces to the nanoscale, the surface-to-volume ratio increases.
- This enhances the number of surface active sites (reaction spots).
- It is the main reason for changes in nanomaterial properties compared to bulk materials.

<u>Optical Properties:</u>

- Optical properties of nanomaterials differ significantly from their bulk counterparts.
- These changes are mainly due to:
  - Surface Plasmon Resonance (SPR)
  - Increased energy level spacing at nanoscale

**Surface Plasmon Resonance (SPR):**

- SPR is the coherent excitation of free electrons in the conduction band of a metal nanoparticle.
- It occurs when the metal nanocrystal size is smaller than the wavelength of the incident light.

- Plasmon represents the collective oscillation of free charge in a metal, considered as a kind of plasma wave.
- In metals, the positive charge is fixed while free electrons can move.
- When an external electric field (e.g., from light) is applied, surface electrons vibrate collectively.
- These vibrating electrons generate their own electric field.
- When the electric field of vibrating electrons resonates with the external field, SPR occurs.

Examples:

- Spherical Au nanoparticles:
    - SPR band at 520–550 nm
    - Absorb green light → appear red-purple
- Colloidal Ag nanoparticles:
    - SPR band near 400 nm
    - Absorb blue light → appear yellow

<u>Catalytic Properties:</u>

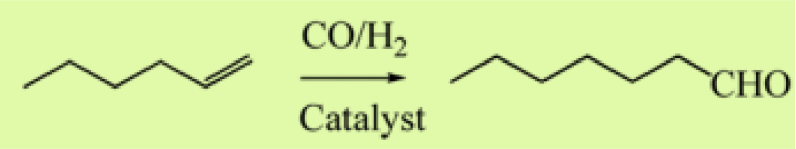
- Nanocatalysts have:
    - High specific surface area
    - High surface energy
- This leads to:
    - Enhanced catalytic activity
    - Higher selectivity
    - Reactions at lower temperatures
    - Fewer side reactions
    - Higher recycling rate and energy efficiency
- Widely used in:
    - Green chemistry
    - Environmental remediation
    - Biomass conversion
    - Renewable energy systems

**Mechanism:**

- Reaction rate depends on the number of active sites on the catalyst.
- Nanoscale catalysts have more active sites than bulk of the same quantity due having larger surface areas.
- Unsaturated valencies of surface atoms make nanoparticles highly reactive.
- Thus, nanocatalysts show higher efficiency compared to bulk materials.

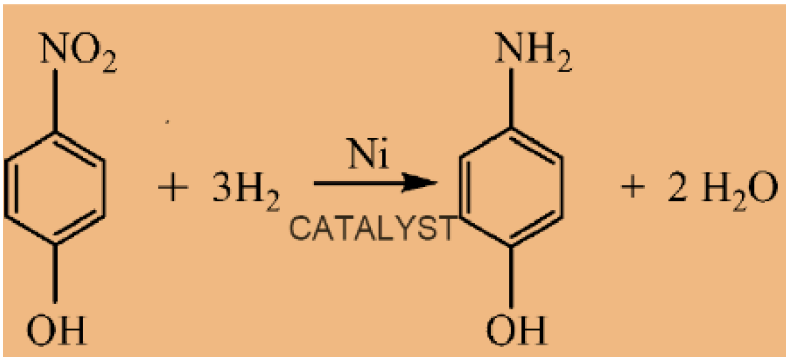
**Examples:**

1. Hydroformylation of 1-Hexene:




- Rh/nano-ZnO catalyst → 100% conversion, 96% aldehyde yield
- Rh on bulk ZnO → only 76% yield but be more stable and active
- Nano-ZnO catalyst is more stable and active

2. Hydrogenation of p-nitrophenol to p-aminophenol:




- Nano-sized Ni → 15× faster than Raney Ni under same conditions
- Deactivation after 6 cycles:
  - Nano Ni: 54%
  - Raney Ni: 80.4%
- Nano Ni shows superior stability and reusability

**Question:**
**Classify nanomaterials based on their dimensions.**

**Answer:**

1. Zero Dimensional (0D): Less than 100 nm in all three dimensions.
2. One Dimensional (1D): Less than 100 nm in any two dimensions.
3. Two Dimensional (2D): Less than 100 nm in any one dimension.
4. Three Dimensional (3D):
   - Aggregations of 0D, 1D, or 2D nanomaterials.
   - Greater than 100 nm in all three dimensions.

**Question:**
**Explain the low temperature solution combustion (SCS) method for the synthesis of nanomaterials. Mention its principle, procedure, and advantages.**

**Answer:**

A low temperature, time-saving, energy-efficient, self-propagating method. Involves a spontaneous exothermic redox reaction between metal nitrate (oxidizer) and organic fuel (reducer). Used to synthesize nano metal oxides.

Principle:

- Heat energy from the exothermic redox reaction between metal nitrate and fuel is used.
- Metal nitrates act as oxidizers.
- Fuels include glycine, oxalic acid, urea, hexamine, sugar, EDTA, dextrose, etc.

Reaction (General Form):
Metal nitrate + Fuel → (500 ºC) → Nano metal oxide + Gases

Fuel Calculation:

- Total oxidizing valency of the fuel must match the total reducing valency of the metal nitrate.
- Use valency values: N = 0, O = -2, H = +1, C = +4, $M^{n+}$ = +n

Procedure:

*Step 1: Weighing and Mixing*

- Weigh metal nitrate and fuel according to stoichiometry.
- Dissolve in minimal distilled water in a beaker.
- Stir the mixture on a magnetic stirrer to get a uniform redox solution.

*Step 2: Evaporation and Combustion*

- Place the clear solution in a furnace maintained at 500 ºC.
- Solution first undergoes evaporation → forms a more concentrated, uniformly mixed viscous gel.
- Later, gel ignites spontaneously in the redox mixture, producing flame or smouldering combustion.
- Reaction completes in 1–2 minutes, releasing gases and forming nano metal oxide.

Advantages of SCS Method:

- Easy and fast process using simple equipment.
- Produces high-purity nanomaterials.
- Allows control over composition, structure, size, and stoichiometry.
- High exothermicity permits doping of industrially useful phosphors.
- Cheaper than conventional solid-state methods.

**Question:**
**Write a brief procedure for the synthesis of MgO by solution combustion method.**

**Answer:**

- Chemicals used: Magnesium nitrate [$Mg(NO_3)_2 \cdot 6H_2O$], urea [$NH_2CONH_2$], distilled water

- Molecular weights: $Mg(NO_3)_2 \cdot 6H_2O$ = 256.41, urea = 60
- Valencies:
    - $Mg(NO_3)_2$ = 10
    - Urea = 6
- Fuel calculation:
    - Wt. of urea = (5 × 10 × 60) / (256.41 × 6) = 1.95 g

Procedure:

- Weigh 5.0 g of magnesium nitrate and 1.95 g of urea.
- Dissolve both in minimum distilled water and stir for 10 minutes to get a uniform redox mixture.
- Place the mixture in a furnace at 500 °C.
- The solution evaporates, forms a viscous gel, then combusts spontaneously.
- A porous white nano MgO powder forms.
- Confirm nano properties using XRD and TEM.

**Question:**
**Explain the sol-gel method for the preparation of nanometal oxides. Mention the reactions, process steps, and drying methods.**

**Answer:**

A wet chemical technique widely used for fabrication of nano-structured ceramic materials and thin films.
Converts precursor solution (metal salts or metal alkoxides) into nanostructured inorganic solids via inorganic polymerization reactions catalyzed by water.
Metal alkoxides (M–OR) are commonly used as precursors due to their reactivity with water.

<u>Reactions Involved:</u>

1. Hydrolysis:
   $M–OR + H_2O \rightarrow M–OH + ROH$
   (Metal alkoxide → Metal hydroxide)

2. Condensation:
   $M–OH + M–OR \rightarrow M–O–M + ROH$
    - Forms a three-dimensional polymer network.
    - M–O–M bonding extends in all directions.

<u>Process Steps:</u>

- **Step 1: Preparation of homogeneous solution**
  Dissolve metal salt in water, or metal alkoxide in an organic solvent (usually alcohol) that is miscible with water.

- **Step 2: 'Sol' formation (Hydrolysis)**
  Homogeneous solution transforms into a sol (stable dispersion of colloidal precursor particles in solvent).

During hydrolysis, nucleophilic attack by oxygen of water replaces –OR groups → releases alcohol → forms metal hydroxide.

- **Step 3: 'Gel' formation (Condensation)**
  The sol is aged to promote condensation between metal hydroxyl/alkoxy species.
  Forms M–O–M bonds and releases $H_2O$ or ROH.
  Results in a rigid, interconnected, porous inorganic network fully saturated with liquid (gel).
  This transition is called the Sol-Gel transition.

- **Step 4: Drying of Gel**
  Removes the liquid phase from the gel network.

  Three types of drying methods:
  a) Supercritical drying → removes solvent under hypercritical conditions without destroying network → forms aerogel
  b) Ambient drying → removes ROH groups under normal conditions → forms xerogel
  c) Sintering at 800°C → causes densification, decomposition, and collapse of gel into powder

<u>Advantages of the sol-gel method:</u>

- simple and economical technique; does not require expensive machinery or apparatus.
- Offers high flexibility to produce materials with a wide range of stoichiometry and dopants.
- Allows fabrication of high-quality coatings.
- Starting materials are easily available, inexpensive, and of high purity.
- Enables preparation of nanomaterials in a variety of shapes.

**Question:**
**Write a brief procedure for the preparation of $TiO_2$ by the sol-gel method.**

**Answer:**

- Chemicals used: Titanium isopropoxide, absolute alcohol, distilled water, $HNO_3$
- Equipment: 250 ml beaker, magnetic stirrer, furnace, oven

<u>Procedure:</u>

- Dissolve titanium isopropoxide in ethanol and water in a 1:4 ratio.
- Add $HNO_3$ to adjust pH and slow hydrolysis.
- Stir the solution vigorously for 30 minutes to form a sol.
- Age the sol for 24 hours → transforms into gel by polycondensation.
- Dry the gel at 120 °C for 2 hours.
- Sinter the dry gel at 450 °C for 2 hours to obtain $TiO_2$ nanoparticles.

<u>Reaction mechanism:</u> Nucleophilic attack by water on the alkoxy group.

**Question:**
**Describe the types, synthesis, and growth mechanism of carbon nanotubes (CNTs).**

**Answer:**

Carbon nanotubes (CNTs) are cylindrical structures formed by rolling a hexagonal network of carbon atoms.

Types of CNTs:

1. Single-Walled Nanotubes (SWNT):
    - Types: Armchair, Zig-Zag, Chiral
2. Multi-Walled Nanotubes (MWNT):
    - Consist of multiple rolled layers of graphene sheets

**Synthesis – Modified Chemical Vapor Deposition (CVD):**

Requirements:

- High temperature tubular furnace (500–1500 °C)
- Carbon source: Benzene, xylene, natural gas, acetylene
- Substrate: Carbon, quartz, silicon
- Inert gas: Argon, Hydrogen, Nitrogen
- Catalyst: Ferrocene, Nickellocene, Cobaltocene

Working Principle:

- Hydrocarbon precursors decompose at 500–1200 °C on catalyst-activated substrate in inert atmosphere.
- Equipped with a horizontal tubular furnace as the reactor. The tube is made of quartz, 30 mm in diameter and 1000 mm in length.
- Ferrocene acts as catalyst, benzene/toluene as carbon source.
- Gases transported by Ar, $H_2$, or both into the quartz tube reactor.
- Decomposition of Fe and carbon produces their ions → form carbon nanostructures.
- Growth occurs in or near the heating zone (500–1150 °C), typically over 30 minutes.

Growth Mechanism of CNTs:

1. Diffusion of hydrocarbon gas into the reactor.
2. Adsorption of hydrocarbon gas on catalyst surface.
3. Homolytic fission (dissociation) of hydrocarbon → forms reactive C and H free radicals.
4. Repetition of 1, 2 & 3 increases concentration of free radicals.
5. Precipitation of carbon radicals on the catalyst surface → tube-like crystal growth. H radicals combine to form $H_2$ gas and escape.

**Question:**
**Explain the functionalization of carbon nanotubes (CNTs) and list their applications.**

**Answer:**

Functionalization of carbon nanotubes (CNTs) is the process of attaching chemical groups to their surface

Need for Functionalization:

- CNTs have poor chemical affinity with organic matrices.
- Dispersion into matrices is a major challenge.
- Functionalization improves compatibility and expands application potential in nanodevices and electronics.

Types of Functionalization:

a) Chemical (Covalent) Functionalization:

- Involves forming covalent bonds between functional groups and the carbon framework of CNTs.
- Can be done at:
    - End caps of nanotubes
    - Sidewalls with defects
- Changes carbon hybridization from $sp^2$ to $sp^3$.
- Causes loss of π-conjugation system.

b) Physical (Non-Covalent) Functionalization:

- Does not alter the conjugated system of CNT sidewalls.
- Preserves structural properties.
- Uses π–π stacking or hydrophobic interactions.
- Common agents:
    - Aromatic compounds
    - Surfactants
    - Polymers
    - Biopolymers
- Improves solubility without damaging structure.
- Examples: Aromatic molecule absorption, polymer wrapping.

Applications of CNTs:

- Optoelectronic devices (e.g., Field Emission Displays)
- Electrodes in batteries, capacitors, and supercapacitors
- Catalyst supports in PEM fuel cells
- Hydrogen storage material in fuel cell vehicles
- Artificial muscles powered by electricity
- Electrocatalyst for water splitting and hydrogen production
- Adsorbents and photocatalysts for degradation of organic pollutants
- Sensors for detecting heavy metals, gases, VOCs, and biomolecules
- Drug delivery carriers

**Question:**
**Describe graphene, its synthesis by Modified Hummer's method, functionalization types, and applications.**

**Answer:**

- An allotrope of carbon made of a single layer of carbon atoms arranged in a hexagonal pattern.
- It is a two-dimensional, honeycomb-structured, one-atom-thick sheet.
- Carbon atoms are bonded via **$sp^2$ hybridization**.
- Purely carbon-based and exhibits unique electrical, mechanical, and optical properties.

<u>Synthesis – Modified Hummer's Method (for Graphene Oxide and Reduced Graphene Oxide):</u>

1. Mix graphite (2 g) and $NaNO_3$ (2 g) with $H_2SO_4$ (90 ml); stir in an ice bath for 30 min.
2. Add $KMnO_4$ (10 g); ultrasonicate at 50 °C for 2 hours.
3. Add DI water (200 ml) and $H_2O_2$ (12 ml, 35%) slowly.
4. Wash with HCl (300 ml, 10%), followed by concentrated HCl (200 ml, 37%).
5. Dry to obtain graphene oxide (GO) powder.
6. Reduce GO by reacting with sodium borohydride to obtain reduced graphene oxide (rGO).

<u>Functionalization of Graphene:</u>

- Same as CNT; improves dispersion and modifies physical/chemical properties.
- Reduces cohesive forces between graphene sheets.

1. Covalent (Chemical) Functionalization:
    - Functional groups form covalent bonds with graphene.
    - Alters the bonding network and modifies properties.
    - Examples: Amine, cysteine, thiolene group attachments.

2. Non-Covalent (Physical) Functionalization:
    - Maintains π-conjugation system.
    - Uses π–π stacking or hydrophobic interactions.
    - Preserves structural and electronic integrity.

<u>Applications of Graphene:</u>

- Energy: Solar cells, batteries, fuel cells
- Medicine:
    - Drug delivery
    - Cancer treatment
    - Gene delivery
    - Photothermal therapy
    - Diabetes monitoring
    - Dialysis

    - Bone and teeth implantation
    - Tissue engineering, cell therapy
- Electronics and Sensors:
    - UV sensors
    - Biosensors
    - Transistors
    - Wearable electronics
    - Flexible screens
    - Optoelectronics
    - Optical sensors
- Water Technology: Water purification, desalination
- Others: Lubricants

**Q: What are thin films? Explain the classification, properties, and uses of thin films.**

A thin film is a layer of material ranging from fractions of a nanometer (monolayer) to several micrometers in thickness.

The controlled synthesis of materials as thin films (a process referred to as deposition) is a fundamental step in many applications.

Thin films are classified into four types:
• Single crystals
• Epitaxial
• Polycrystalline
• Amorphous

The properties of thin films are as follows:
• High surface to volume ratio
• Geometrical control (dictated by substrate)
• Compact
• Single crystal-like properties

Thin films are used in the following applications:
• Microelectronic devices
• Telecommunication devices
• Decorative coatings
• Optical coatings (windows, solar cells, etc.)
• Sensors
• Catalysts

Thin Film Deposition can be achieved through two methods: Physical Vapour Deposition (PVD) or Chemical Vapour Deposition (CVD).

**Q: Explain Chemical Vapour Deposition (CVD), its apparatus, steps involved, advantages, disadvantages, and applications.**

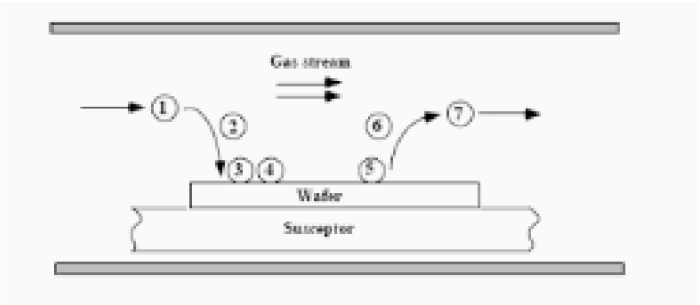
Chemical Vapour Deposition (CVD) is the formation of a non-volatile solid film on a substrate by the reaction of vapor phase chemicals (reactants) that contain the required constituents.

The reactant gases are introduced into a reaction chamber and are decomposed and reacted at a heated surface to form the thin film.

CVD Apparatus

- Gas delivery system – For the supply of precursors to the reactor chamber.
- Reactor chamber – Chamber within which deposition takes place.
- Substrate loading mechanism – A system for introducing and removing substrates, mandrels etc.
- Energy source – Provide the energy/heat that is required to get the precursors to react/decompose.
- Vacuum system – A system for removal of all other gaseous species other than those required for the reaction/deposition.
- Exhaust system – System for removal of volatile by-products from the reaction chamber.
- Exhaust treatment systems – In some instances, exhaust gases may not be suitable for release into the atmosphere and may require treatment or conversion to safe/harmless compounds.
- Process control equipment – Gauges, controls etc. to monitor process parameters such as pressure, temperature and time. Alarms and safety devices would also be included in this category.

Steps involved in a CVD process




1. Transport of reactants by forced convection to the deposition region.
2. Transport of reactants by diffusion from the main gas stream through the boundary layer to the wafer surface.
3. Adsorption of reactants on the wafer surface.
4. Surface processes, including chemical decomposition or reaction, surface migration to attachment sites (such as atomic-level ledges and kinks), site incorporation, and other surface reactions.
5. Desorption of byproducts from the surface.
6. Transport of byproducts by diffusion through the boundary layer and back to the main gas stream.

7. Transport of byproducts by forced convection away from the deposition region.

Advantages

- High growth rates possible.
- Can deposit materials which are hard to evaporate.
- Good reproducibility.
- Can grow epitaxial films.

Disadvantages

- High temperatures.
- Complex processes.
- Toxic and corrosive gases.

Applications

- Coatings such as:
  - Wear resistance
  - Corrosion resistance
  - High temperature protection
  - Erosion protection
  - Combinations thereof
- Semiconductors and related devices, such as: ICs & Sensors

**Q: Explain Plasma-Enhanced Chemical Vapor Deposition (PECVD), its working principle, advantages over conventional CVD, and its significance in thin film deposition.**

Plasma-enhanced chemical vapor deposition (PECVD) is a chemical vapor deposition process used to deposit thin films from a gas state (vapor) to a solid state on a substrate. Chemical reactions are involved in the process, which occur after creation of a plasma of the reacting gases.
The plasma is generally created by radio frequency (RF) (alternating current (AC)) frequency or direct current (DC) discharge between two electrodes, the space between which is filled with the reacting gases.

Advantages of PECVD over conventional CVD

- Lower deposition temperatures.
- Good conformity and step coverage on uneven surfaces.
- Tighter control of the thin film process.
- High deposition rates.

Temperature Comparison and Benefits

- Standard CVD temperatures are usually conducted in 600°C to 800°C.
- PECVD temperatures range from room temperature to 350°C.

- Lower temperature enables successful applications in situations where the higher CVD temperatures could potentially damage the device or substrate being coated.
- Operating at a lower temperature creates less stress between thin film layers that have different thermal expansion/contraction coefficients.
- This allows for high-efficiency electrical performance and bonding to very high standards.

Conformity and Coverage

- CVD is a diffusive gas-driven process, allowing better coverage on uneven surfaces such as trenches or walls with relatively high conformity.
- PVD coatings are line-of-sight deposition, which can result in more variance in thin film depth if irregularities shield areas from coating.
- PECVD reduces the line-of-sight issue for high conformity of thin films because the plasma stream can surround the substrate.

Plasma Parameter Control

- By varying the parameters of the plasma, excellent control of the material properties can be achieved including:
  - Density and hardness
  - Purity
  - Roughness
  - Refractive index of optical films
- PECVD yields some of the fastest deposition rates while still maintaining high film quality.

Unique Capabilities

- PECVD can produce unique compounds and films that cannot be created by common CVD techniques alone.
- It can produce films that demonstrate very high solvent and corrosion resistance, along with chemical and thermal stability.

**Q: Explain how Plasma Enhanced Chemical Vapor Deposition (PECVD) works.**

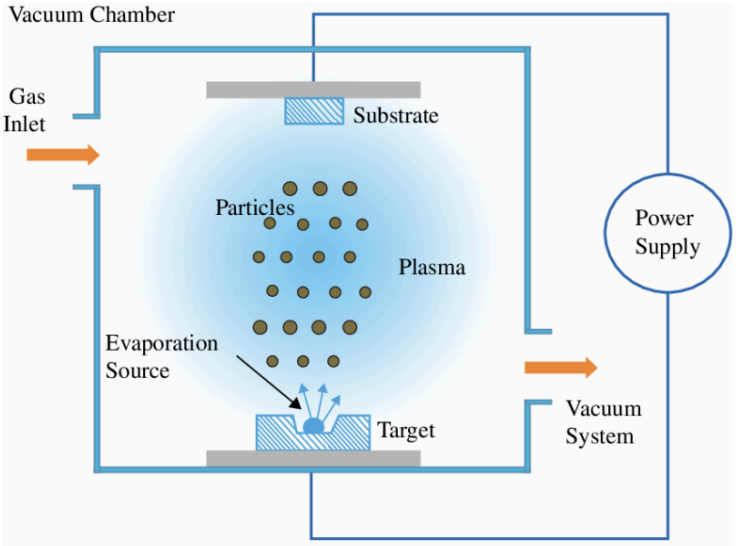
- For typical semiconductor applications, the substrate is placed in the deposition chamber between two parallel electrodes — a grounded electrode and usually an RF-energized electrode.

- The substrate is heated in the range of 250°–350°C.

- Precursor gases such as silane ($SiH_4$) and ammonia ($NH_3$) are typically mixed with inert gases such as argon (Ar) or nitrogen ($N_2$) to control processes.

- These gases are introduced into the chamber via a shower head fixture over the substrate that helps to spread the gas more evenly onto the substrate.

- The plasma is ignited by electrical discharge (100–300 eV) between the electrodes, creating a glowing sheath around the substrate that contributes to thermal energy driving the chemical reactions.
- These reactions first occur in the plasma as a result of precursor gas molecules colliding with highly energized electrons, which then, via gas flow, travel to the substrate where they react and are absorbed on the substrate surface to grow films.
- The chemical by-products are then desorbed and pumped away, completing the deposition process.
- Higher gas flow rates can yield higher deposition rates, which along with operating temperatures, control properties such as thickness, hardness, or refractive index.




*Reactor design for PVD*

**Q: Explain Metal-Organic Chemical Vapour Deposition (MOCVD), its working principle, and applications.**

MOCVD is a technique that is used to grow/deposit thin solid films, usually semiconductors, on solid substrates (wafers) using organometallic compounds as sources.

The films grown by MOCVD are mainly used for the fabrication of electronic and optoelectronic devices.

The electronic and optoelectronic devices produced by MOCVD are used in:
- Cell phones
- Optical communication
- Optical storage (CD, DVD)
- Traffic lights

• Billboards (LEDs)
• Lighting
• Solar cells

Using MOCVD, we can build up many layers, each of a precisely controlled thickness, to create a material which has specific optical and electrical properties.

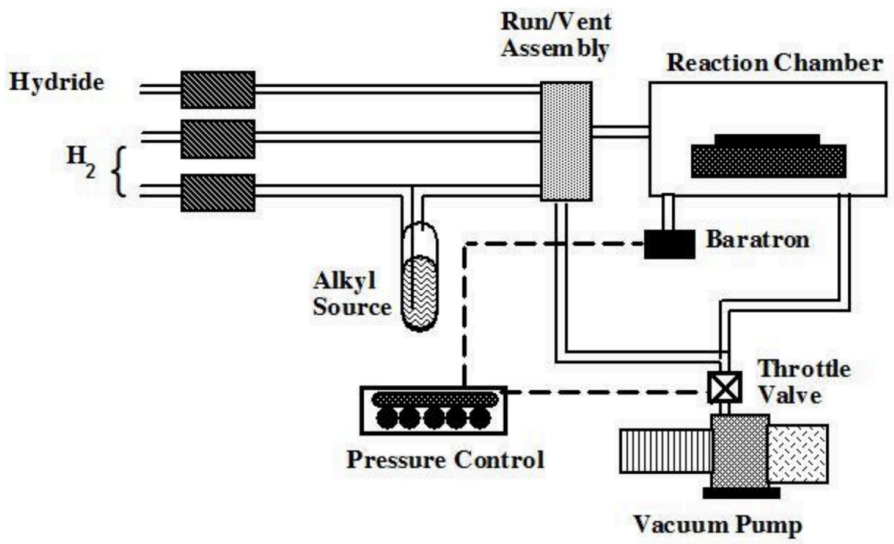
Principle of Operation

- Transport of precursor molecules (Group-III metalorganics + Group-V hydrides or alkyls) by a carrier gas ($H_2$, $N_2$) onto a heated substrate.
- Surface chemical reactions occur to form the thin film.

Applications

- Laser diode
- Transistors
- LED
- Solar Cells

**Q: Describe the main parts of an MOCVD system, the chemical reactions involved in film formation, and the transport and growth mechanisms.**




1. Source Supply System

The different sources which are used are:
• TMI (Trimethylindium) and TEG (Triethylgallium) are used as source material for In and Ga respectively.
• $NH_3$ is used as a source material for nitrogen.
• $N_2$ is used as a carrier gas to bring TMI and TEG into the reactor.
• $H_2$ gas is used for the thermal treatment of the substrate.
• Reactive gases are fed into the reactor through the mass flow controllers (MFC).

2. Reaction Chamber / Reactor

- Epitaxial vapor growth is made inside the reactor at different conditions (temperature, pressure, gas flow).
- The substrates are placed on the susceptor.
- Reactive gases are then fed into the reactor, and these gases react on the substrate and form a grown film.
- The heating method of this reactor is RF induction heating.
- The deposition reaction, which takes place inside the reactor for the growth of the nitride film, is known as pyrolysis.

<u>Chemical Reactions (Organometal Vapor Phase Epitaxy)</u>

The formation of solid InN and GaN film on the substrate happens by the following chemical reactions in vapor phase with the organometallic reactant species. For this reason, it is called organometal vapor phase epitaxy:

• $(CH_3)_3In + NH_3 \rightarrow 3CH_4(v) + InN(s)$
*Trimethylindium gas + Ammonia gas → Methane gas (volatile) + Indium Nitride*

• $(C_2H_5)_3Ga + NH_3 \rightarrow 3C_2H_6(v) + GaN(s)$
*Triethylgallium gas + Ammonia gas → Ethane gas (volatile) + Gallium Nitride*

*[solid film]*

3. Exhaust System

The exhaust system consists of a rotary and diffusion pump.
When reactions take place, the exhaust gases are released to air through the rotary pump and exhaust fan.

<u>Transport and Growth Mechanisms – MOCVD System</u>

- The deposition process takes place on the substrates (wafers).
- The material source should be:
  • Sufficiently volatile
  • Have high enough partial pressure to get good growth rates
  • Stable at room temperature
  • Produce desired element on substrate with easily removable by-products
- Growth of III–V semiconductors:
  • Group III: Generally metalorganic molecules (trimethyl- or triethyl- species)
  • Group V: Generally toxic hydrides (e.g., $AsH_3$, $PH_3$ — flammable as well); alternative: alkyls (TBAs, TBP)